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Repurposing Stable Diffusion Attention for Training-Free Unsupervised Interactive Segmentation

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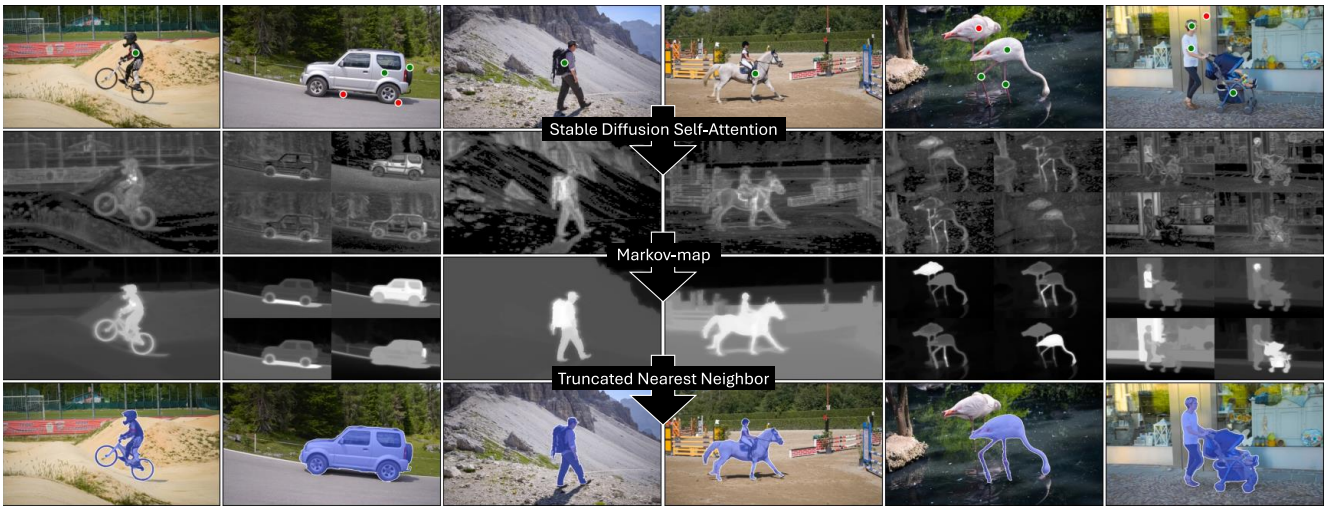


Figure 1. **We introduce M2N2, an unsupervised training-free point prompt based segmentation framework.** We enhance the semantic information present in the self-attention of Stable Diffusion 2 by using a Markov process to generate semantically rich Markov-maps. We then perform a truncated nearest neighbor of each point's Markov-map to obtain a final segmentation.

Abstract

Recent progress in interactive point prompt based Image Segmentation allows to significantly reduce the manual effort to obtain high quality semantic labels. State-of-the-art unsupervised methods use self-supervised pre-trained models to obtain pseudo-labels which are used in training a prompt-based segmentation model. In this paper, we propose a novel unsupervised and training-free approach based solely on the self-attention of Stable Diffusion. We interpret the self-attention tensor as a Markov transition operator, which enables us to iteratively construct a Markov chain. Pixel-wise counting of the required number of iterations along the Markov chain to reach a relative probability threshold yields a Markov-iteration-map, which we simply call a Markov-map. Compared to the raw attention maps, we show that our proposed Markov-map has less noise, sharper semantic boundaries and more uniform values within semantically similar regions. We integrate the Markov-map in a simple yet effective truncated nearest neighbor framework to obtain interactive point prompt based segmentation. Despite being training-free, we experimentally show that our approach yields excellent results in terms of Number of Clicks (NoC), even outperforming state-of-the-art training based unsupervised methods in most of the datasets. Code is available at https://github.com/mkarmann/m2n2.

1. Introduction

The goal of point prompt based interactive image segmentation is to obtain a high-quality segmentation from limited user interaction in the form of clicking. Prompt-based image segmentation gained popularity lately due to large supervised foundation models [18, 35]. In this paper, we focus on unsupervised methods, where no segmentation labels are used at all in the design and/or training of the models. Most recent approaches rely on self supervised backbones like ViT [7], trained either by DINO [4] or MAE [13] based self-

supervised techniques. On the other hand, Stable Diffusion (SD) [36] has been used for many different computer vision applications such as monocular depth estimation [16], semantic segmentation [45], object detection [5] and image classification [20], often resulting in state-of-the-art solutions.

Inspired by DiffSeg [45], we investigate the potential of SD's self-attention maps for training-free interactive segmentation. Generally, for all self-supervised backbones, the main challenges are that attention maps don't distinguish between instances, exhibit noise and are typically of low resolution. We choose SD, particularly version 2 (SD2), as the backbone of our approach, as it provides the highest attention resolution (Sec. 14). To overcome the mentioned challenges, we interpret the self-attention tensors as a Markov transition operator, where the repeated application of the transition forms a Markov chain. We propose a novel Markov-iteration-map or simply Markov-map, where each pixel counts the number of iterations required to obtain a specific probability value. We show that the proposed Markov-map has less noise, the semantic boundaries are sharper and the semantic regions within Markov-maps are more uniformly distributed. Native Markov-maps do not distinguish between instances. Therefore, we further improve Markov-maps with a flood fill approach, which suppresses local minima, to enable instance based segmentation. Finally, we obtain Markov-maps of each prompt point and combine them with a truncated nearest neighbor approach to enable multi-prompt point interactive segmentation. Surprisingly, despite being training-free, we significantly improve the state-of-the-art in terms of Number of Clicks (NoC) and even surpass training based unsupervised approaches in most of the datasets. Our main contributions are:

- We introduce **M**arkov-**M**ap **N**earest **N**eighbor (**M2N2**), the first attention-based unsupervised training-free multi-point prompt segmentation framework.
- We propose a novel method to refine semantic information in attention maps, which we call **Markov-maps**.
- We enable instance aware Markov-maps by utilizing a modified flood fill approach.
- We introduce a truncated nearest neighbor approach to combine multiple point prompts.
- We conduct extensive experiments with multiple backbones and achieve state-of-the-art results, surpassing even unsupervised training-based methods.

2. Related Work

Point prompt based interactive image segmentation has been approached from multiple perspectives. In this paper, we distinguish between supervised and unsupervised methods.

2.1. Supervised Methods

In [52] a click map and click sampling strategies are used in combination with FCN [29], creating the foundation for many follow-up methods. More recent approaches include [43], [24], [17, 18], [34], [15, 42], [27], [54], [6, 25], [26], [44], [28], [14], [46], either deploying dense fusion, where prompts are encoded as prompt maps, or sparse fusion, where prompts are transformed into embedding space to be fused with image and other related embeddings.

Although the supervised approaches achieve good performance and efficiency, they require large-scale pixel-level annotations to train, which are expensive and laborious to obtain. While many supervised methods are tested on additional domains like medical images, it is unclear if there are other domains where the trained models would have a domain gap.

2.2. Unsupervised methods

Classical, unsupervised methods not based on Deep Learning like GraphCut [38], Random Walk [8], Geodesic Matting [2], GSC [9] and ESC [9] have been proposed. However, recently Deep Learning based approaches show great potential. Especially, methods utilizing self-supervised learning achieve impressive results. Such methods rely on pre-trained models (e.g., DenseCL [47], DINO [4]) to extract segments from their features. In [40] some heuristics are proposed to choose pixels belonging to the same object according to their feature similarity. [50] introduces normalized cuts [39] on the affinity graph constructed by pixel-level representations from DINO to divide the foreground and background of an image. In [11] a segmentation head is trained by distilling the feature correspondences from DINO. [31] adopts spectral decomposition on the affinity graph to discover meaningful parts in an image. FreeSOLO [48] designs pseudo instance mask generation based on multi-scale feature correspondences from densely pre-trained models and trains an instance segmentation model with these pseudo masks.

Recently, several papers used Stable Diffusion (SD) [36] for various kinds of applications targeting unsupervised Semantic Segmentation. In [45] the self-attention maps of SD were used with KL-Divergence based similarity measures to merge semantically similar regions in order to extract segments. MIS [21] and UnSAM [49] use methods to capture the semantic hierarchy and create pseudo labels, which are used in a follow-up training process to obtain a model for promptable segmentation. UnSAM additionally allows automatic whole-image segmentation, but does not report NoC values on any dataset. In [5] object detection is formulated as a denoising diffusion process from noisy boxes to object boxes, and [22] proposes training-free unsupervised segmentation using a pre-trained diffusion model by iteratively contrasting the original image and a painted image

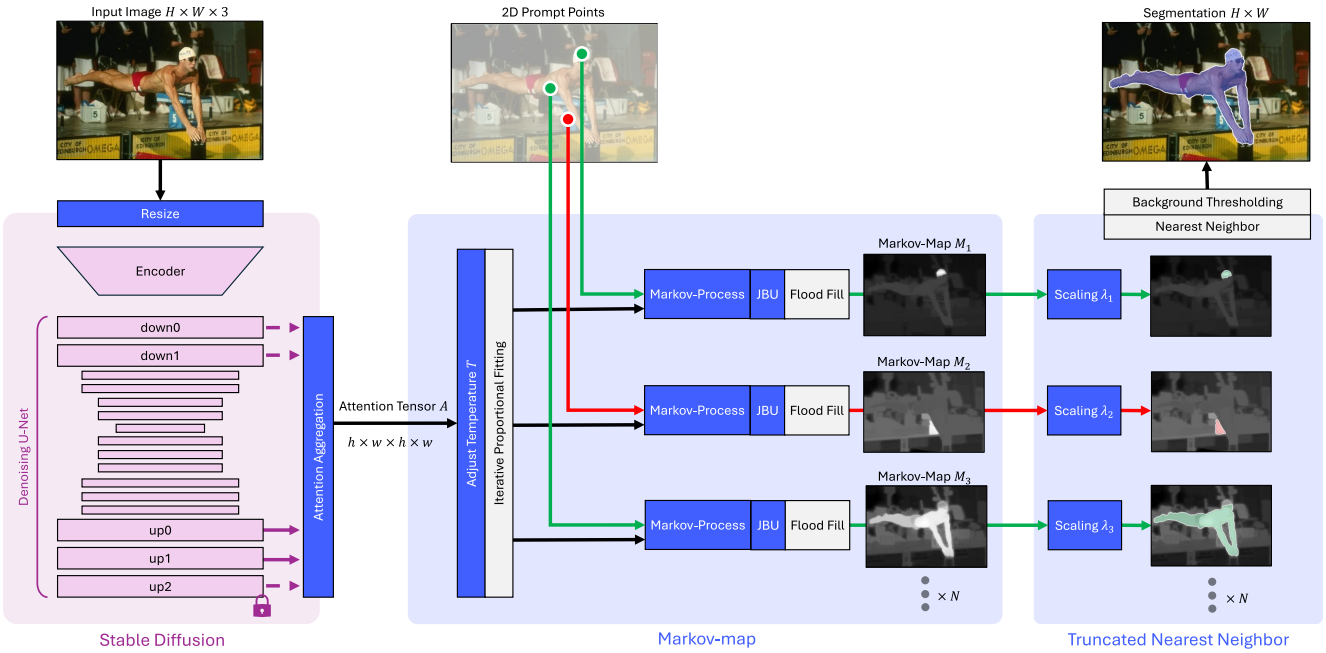


Figure 2. **M2N2 framework overview**. We perform a single denoising step of the input image with Stable Diffusion 2 to obtain attention tensors. The tensors are aggregated and utilized to obtain a Markov-map M_i for each prompt point. The final segmentation is the result of a truncated nearest neighbor of scaled Markov-maps M_i as a measure of semantic distance for each prompt point. The green and red areas in the scaled Markov-maps denote regions where the distance is less or equal to the global background threshold. In this visualization, components in blue contain adjustable hyperparameters.

in which a masked area is re-painted. While this method obtains good results, it does not provide multi-prompt interactive segmentation.

Inspired by [45], we focus on SD's self-attention maps as an initial semantic feature map. Our proposed Markov-map improves the semantic features, resulting in an excellent, unsupervised point prompt based interactive segmentation solution.

3. Method

In this section, we introduce our **M**arkov-**M**ap **N**earest **N**eighbor (M2N2) framework in full detail. We design our interactive process similar to [27][18] where the user is shown an image with a suggested (initially empty) segmentation and allowing them to place a foreground or background point on objects they want to either include or exclude from the segmentation. Once a new point is set, M2N2 uses the entire set of points to predict a new segmentation. This interactive process continues until a satisfactory segmentation is achieved. Fig. 2 shows the three main stages of M2N2 applied on an example in which three points have already been set. In the first stage, we obtain an attention tensor of the input image by aggregating SD's self-attentions. The second stage extracts and enhances the semantic information of the attention tensor by creating a Markov-map for each prompt point. The semantically rich Markov-maps are then utilized in a truncated nearest neighbor algorithm to obtain a training-free unsupervised segmentation. In the following subsections, we first formulate the problem in the context of a nearest neighbor algorithm in Sec. 3.1, followed by an explanation of the acquisition of the attention tensor in Sec. 3.2 and extraction of the Markov-maps including the flood fill approach in Sec. 3.3. Finally, we introduce the full M2N2 algorithm Sec. 3.4.

3.1. Truncated Nearest Neighbor

In point prompt based segmentation we are given an image $I \in \mathbb{R}^{H \times W \times 3}$ of width W and height H and a set of labeled prompt points $D = \{(x_1, y_1), (x_2, y_2), ..., (x_N, y_N)\}$ where $x_i \in \mathbb{R}^2$ are the 2D spatial coordinates of each prompt point in image pixel space and the labels $y_i \in \{0, 1\}$ denote whether a point x_i belongs to the background $y_i = 0$ or foreground $y_i = 1$. To perform k-NN segmentation with $k = 1$, we assign each query pixel x_q of our output segmentation the class of its nearest neighbor x_{i^*}:

$$i^* = \arg\min_i d(x_i, x_q), \qquad y_q = y_{i^*} \tag{1}$$

where $d(\cdot, \cdot) : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}_{\geq 0}$ is a semantic distance function measuring the semantic similarity between the query pixel x_q and prompt point x_i. In our case, we allow $d(\cdot, \cdot)$

to be asymmetric, meaning that $d(x_i, x_q)$ may, but does not have to, equal $d(x_q, x_i)$. Using a canonical 1-NN, a single foreground prompt point would always segment the entire image as foreground, requiring a minimum of two prompts, one foreground and one background prompt, to get a useful segmentation. To mitigate this limitation and reduce NoC, we further extend the 1-NN algorithm with a distance threshold of 1 and define $\hat{y}_q$ as:

$$\hat{y}_q = \begin{cases} y_{i^*} & \text{if } d(x_{i^*}, x_q) \leq 1 \\ 0 & \text{otherwise.} \end{cases} \tag{2}$$

If the distance between a query pixel x_q and its nearest neighbor x_{i^*} exceeds the threshold of 1, it is classified as background $\hat{y}_q = 0$, independent of its nearest neighbor's class y_{i^*}. The main challenge is to find a good distance function $d(\cdot, \cdot)$ which measures the semantic dissimilarity of a prompt point x_i and a query pixel x_q.

3.2. Attention Aggregation

In this paper, we use the pre-trained SD 2. Given an image I, we perform a single denoising step by computing a forward pass through the denoising U-Net and extract the multi-head self-attentions $\boldsymbol{S}_j$ of each transformer block j. Each tensor $\boldsymbol{S}_j$ is of the shape $N_{heads} \times h \times w \times h \times w$. N_{heads} denotes the number of attention heads and h and w are the height and width of the attention maps respectively. Each attention map $\boldsymbol{S}_j[n, k, l, :, :]$ of the tensor $\boldsymbol{S}_j$ is a probability distribution, meaning the sum of each map's elements is equal to 1. We define the aggregated tensor $\boldsymbol{A} \in \mathbb{R}^{h \times w \times h \times w}$ as follows[1]:

$$\boldsymbol{A}[:, :, :, :] = \sum_{j \in \text{AttnBlocks}} w_j \sum_{n}^{N_{heads}} \boldsymbol{S}_j[n, :, :, :, :] \tag{3}$$

Ensuring that the sum of weights $w_j \in [0, 1]$ is 1, the result is a single attention tensor $\boldsymbol{A}$ of which each 2D-map $\boldsymbol{A}[k, l, :, :]$ is a probability distribution.

3.3. From Attention Tensor to Markov-Maps

First, we flatten the attention tensor $\boldsymbol{A} \in \mathbb{R}^{h \times w \times h \times w}$ to obtain a matrix $A \in \mathbb{R}^{(h \cdot w) \times (h \cdot w)}$. Since A is a right stochastic matrix[2], we can use it as a transition matrix in a Markov chain:

$$p_t = p_0 \cdot A^t \quad \text{with} \quad p_0 = onehot(x) \tag{4}$$

where the probability distribution row vector $p_0 \in \mathbb{R}^{1 \times (h \cdot w)}$ is the start state and $p_t \in \mathbb{R}^{1 \times (h \cdot w)}$ is the state after $t \in \mathbb{Z}_{\geq 0}$ iterations. The start state p_0 is the one-hot encoded vector of a prompt point x. A is strictly positive and generally irreducible, aperiodic stochastic matrix. Therefore, for $t \to \infty$, each p_0 converges to a stationary distribution p_∞ of the Markov chain [37]. The state p_∞ depends on the attention matrix A and is therefore different for each image. In order to be image agnostic, we remove the per-image bias in p_∞, by applying iterative proportional fitting (IPF) to the matrix A. This converts A to a doubly stochastic matrix. For all images, A has therefore the property that each start state p_0 converges to a uniform distribution $p_\infty = \frac{1}{h \cdot w}\mathbf{1}$ for $t \to \infty$ [41]. As a result, the Markov chain describes the process from the lowest possible entropy in p_0 to the highest possible entropy in p_∞[3]. Importantly, we are able to control the rate of convergence by changing the entropy of the attention maps $A[k, :]$ by modifying the temperature $T \in \mathbb{R}_{>0}$:

$$A_{new}[k, l] = \frac{\exp\left(\frac{1}{T} \log(A_{prev}[k, l])\right)}{\sum_j \exp\left(\frac{1}{T} \log(A_{prev}[k, j])\right)} \tag{5}$$

Eq. (5) is applied before the IPF to ensure that the transition matrix A is doubly stochastic. Temperatures T higher than 1 increase the overall entropy of A and therefore require fewer time steps t to reach a uniform distribution in p_t. The key idea is to measure the time t it takes for each element k in p_0 to converge to the uniform distribution:

$$m[k] = \min\left\{t \in \mathbb{Z}_{\geq 0} \mid \frac{p_t[k]}{\max p_t} > \tau\right\} \tag{6}$$

Here $\max p_t$ returns the maximum element of the vector p_t and the vector $m \in \mathbb{Z}_{\geq 0}^{1 \times (h \cdot w)}$ stores the minimal time required of each element k to saturate, i.e., to reach the relative probability threshold τ. As a result, the hyperparameter τ allows us to control how quickly the element k saturates. Since m only contains integers, we also perform a linear interpolation between consecutive time frames to obtain a smoother result. We then reshape m into a matrix of the shape $h \times w$ and upscale it with JBU [19] to the input image resolution of $H \times W$, resulting in $M \in \mathbb{R}^{H \times W}$. Since M measures the number of iterations of the Markov chain, we call the resulting map the Markov-iteration-map or in short Markov-map.

Flood Fill. The task of point prompt based segmentation typically requires instance segmentation. This proves to be challenging for the proposed Markov-maps, since self-attentions do not take object instances into account.

[1] We do not require any resizing or upscaling since we only consider the 5 attention tensors illustrated in Fig. 2. These tensors are of the highest resolution in the U-Net, providing attention tensors $\boldsymbol{S}_j$ of the same shape. We simply average along the attention-head dimension with equal weights since we assume all heads represent very similar attentions [45].

[2] This is the case because each attention map $\boldsymbol{A}[i, j, :, :]$ is a probability distribution and so is $A[k, :]$

[3] The one-hot encoded start state has one element set to 1 and all others set to 0 and therefore an entropy of 0. The uniform distribution in p_∞ yields the maximum uncertainty and therefore the largest entropy.

Attention Aggregation Weights					DAVIS	
w_{down0}	w_{down1}	w_{up0}	w_{up1}	w_{up2}	NoC85↓	NoC90↓
1	0	0	0	0	11.34	15.25
0	1	0	0	0	9.38	13.18
0	0	1	0	0	4.86	<u>6.90</u>
0	0	0	1	0	<u>4.79</u>	7.10
0	0	0	0	1	6.00	9.02
0	0	0.5	0.5	0	**4.60**	**6.72**

Table 1. **Ablation study of the attention blocks** on DAVIS. The blocks up0 and up1 achieve the lowest individual NoC.

Score Functions $s_{i,\cdot}(\lambda)$				DAVIS	
$s_{i,\text{prior}}$	$s_{i,\text{edge}}$	$s_{i,\text{pos}}$	$s_{i,\text{neg}}$	NoC85↓	NoC90↓
×	×	×	×	6.47	10.29
×	✓	✓	✓	<u>4.67</u>	<u>6.81</u>
✓	×	✓	✓	7.20	9.99
✓	✓	×	✓	4.92	6.87
✓	✓	✓	×	5.46	8.66
✓	✓	✓	✓	**4.60**	**6.72**

Table 2. **Ablation study of threshold score functions** $s_{i,\cdot}(\lambda)$ on DAVIS. The first row utilizes no score with $\lambda_i = 0.5$.

Therefore, we propose a modified flood fill approach. Given a prompt point x and its corresponding Markov-map M, we perform a flood fill on M, using the pixel at x as the starting pixel. In contrast to classical flood fill, we do not set each pixel to a desired color. Instead, we store the minimum required flood threshold to reach each pixel. This simple yet effective approach suppresses local minima and ensures a global minimum at the prompt point x. Our approach requires that the instances do not overlap[4]. After applying our modified flood fill we obtain the final Markov-map.

3.4. M2N2: Markov-Map Nearest Neighbor

Using the one-hot encoded coordinates of each prompt point x_i as start states, we generate a Markov-map M_i for each prompt point x_i. This allows us to construct the following distance function for the truncated nearest neighbor:

$$d(x_i, x_q) = \frac{M_i[x_q]}{\lambda_i} \tag{7}$$

where $M_i[x_q]$ denotes the value of the Markov-map M_i at the pixel x_q. Since every distance greater than 1 is assigned as background, we can use the scalar λ_i as a threshold to truncate the Markov-map M_i. Choosing a good threshold λ_i is crucial to reduce the NoC. Therefore, we introduce a heuristic to adaptively determine an optimal λ_i:

$$\lambda_i = \arg\max_{\lambda} s_i(\lambda) \tag{8}$$

Given a threshold λ, we define the total score function $s_i(\lambda)$ as a product of four specific score functions, taking into account a prior, semantic edges and prompt point consistencies:

$$s_i(\lambda) = s_{i,\text{prior}}(\lambda) \cdot s_{i,\text{edge}}(\lambda) \cdot s_{i,\text{pos}}(\lambda) \cdot s_{i,\text{neg}}(\lambda) \tag{9}$$

Each specific score $s_{i,\cdot}(\lambda)$ is based on the individual segmentation at the threshold λ of a single prompt point x_i:

[4]Please note that M2N2 is still able to separate overlapping instances by utilizing multiple prompts.

- $s_{i,\text{prior}}(\lambda) = 1$ if the segment size relative to the image size is less than 40%, 0 otherwise
- $s_{i,\text{edge}}(\lambda)$ is the average Sobel gradient of M_i computed along the segment boundary pixels
- $s_{i,\text{pos}}(\lambda)$ is the percentage of all prompt points x_j having the same class $y_i = y_j$ inside the segment
- $s_{i,\text{neg}}(\lambda) = 0$ if any other prompt point x_j of wrong class $y_i \neq y_j$ is inside the segment, 1 otherwise.

By inserting Eq. (7) into Eq. (1) and Eq. (2), we arrive at the final M2N2 equations:

$$i^* = \arg\min_i \frac{M_i[x_q]}{\arg\max_{\lambda} s_i(\lambda)} \tag{10}$$

$$\hat{y}_q = \begin{cases} y_{i^*} & \text{if } \frac{M_{i^*}[x_q]}{\arg\max_{\lambda} s_{i^*}(\lambda)} \leq 1 \\ 0 & \text{otherwise} \end{cases} \tag{11}$$

4. Experiments

Datasets. We evaluate M2N2 on 4 public datasets:

- **GrabCut** [38]: 50 images; 50 instances.
- **Berkeley** [30]: 96 images; 100 instances. Some of the images are shared with the GrabCut dataset.
- **SBD** [12]: 2857 images; 6671 instances (validation set).
- **DAVIS** [33]: 345 images; 345 instances. The segmentation is of very high-quality and instances include many objects with fine details and small structures. We use the same 345 images as used in [21, 27].

Evaluation Metrics. Following previous work [21, 27], we evaluate our approach by simulating user interaction, in which we place the next click in the center of the largest error region. The maximum number of clicks for each instance is 20 and we provide two metrics, the average number of clicks required to reach an IoU of 85% as NoC85 and an IoU of 90% as NoC90.

Comparison with Previous Work. Tab. 3 compares our approach M2N2 with previous supervised and unsupervised approaches. To the best of our knowledge, M2N2 is

Method	Backbone	GrabCut [38]		Berkeley [30]		SBD [12]		DAVIS [33]	
		NoC85↓	NoC90↓	NoC85↓	NoC90↓	NoC85↓	NoC90↓	NoC85↓	NoC90↓
Supervised: Trained on Ground Truth Labels									
SimpleClick [27]	ViT-B	1.40	1.54	1.44	2.46	3.28	5.24	4.10	5.48
SimpleClick	ViT-H	**1.32**	1.44	**1.36**	2.09	**2.51**	**4.15**	4.20	5.34
CPlot [26]	ViT-B	1.34	1.48	1.40	2.18	3.05	4.95	**4.00**	5.29
CFR-ICL [44]	ViT-H	-	1.42	-	**1.74**	-	4.45	-	4.77
FocalClick [6]	SegFormerB0-S2	-	1.90	-	3.14	4.34	6.51	-	7.06
InterFormer-Tiny [14]	ViT-L	-	**1.36**	-	2.53	3.25	5.51	-	5.21
OIS [46]	ViT-B	-	-	-	-	-	-	-	**3.80**
Unsupervised: Trained on Pseudo Labels									
TokenCut⋆ [51]	ViT-B	3.04	5.74	6.30	9.97	10.82	13.16	10.56	15.01
FreeMask⋆ [53]	ViT-B	4.06	6.10	5.55	9.02	8.61	12.01	8.26	13.05
DSM⋆ [31]	ViT-B	3.64	4.64	5.49	7.75	8.59	11.57	7.08	10.11
MIS [21]	ViT-B	1.94	2.32	3.09	4.58	**6.91**	**9.51**	6.33	8.44
Unsupervised: Training-Free									
GraphCut [38]	N/A	7.98	10.00	-	14.22	13.60	15.96	15.13	17.41
Random Walk [8]	N/A	11.36	13.77	-	14.02	12.22	15.04	16.71	18.31
Geodesic Matting [2]	N/A	13.32	14.57	-	15.96	15.36	17.60	18.59	19.50
GSC [9]	N/A	7.10	9.12	-	12.57	12.69	15.31	15.35	17.52
ESC [9]	N/A	7.24	9.20	-	12.11	12.21	14.86	15.41	17.70
Attention-NN	SD2	5.48	7.16	6.45	9.61	11.67	14.97	9.66	13.12
KL-NN	SD2	3.80	5.32	5.11	7.71	10.78	14.00	7.60	10.89
M2N2 w/o flood fill	SD2	3.06	4.50	4.80	6.51	9.16	12.26	5.50	7.42
M2N2	ViT-B	1.92	3.40	3.55	5.68	7.94	11.53	6.85	10.86
M2N2	SD1.1	1.84	2.20	2.90	4.93	8.56	11.66	5.16	7.80
M2N2 (Ours)	SD2	**1.62**	**1.90**	**2.45**	**3.88**	7.72	10.94	**4.60**	**6.72**

Table 3. **Main comparison** of M2N2 with baselines and previous work on all four datasets. ⋆ indicates pseudo label generation methods used on SBD images for training an unsupervised SimpleClick model [21]. All supervised methods were trained on SBD, except InterFormer (COCO [23]+LVIS [10]) and OIS (HQSeg44K [17]). Among the unsupervised methods, M2N2 obtains the best results in three out of four datasets, despite being training-free.

the first unsupervised interactive point prompt based segmentation framework utilizing a pre-trained model without requiring any additional training. All other methods either are not based on any deep learning, e.g., GrabCut and related ones, or require the generation of pseudo-labels to train an interactive model, e.g., MIS. Our method surpasses the previous state-of-the-art unsupervised method MIS, which is trained on pseudo labels, on both metrics in three out of four test datasets. We observe the largest improvement on the DAVIS dataset, where we reduced the NoC85 by 1.73 and the NoC90 by 1.72 clicks. We achieve second best results in SBD. A possible explanation for this is that all deep-learning models listed in Tab. 3 are trained on the training set of SBD and therefore might have an advantage on this dataset.

Baselines. We provide two additional baselines in Tab. 3 which use the same framework as M2N2 but without Markov-maps. Attention Truncated Nearest Neighbor (Attention-NN) uses attention maps as a semantic distance measure. KL-Divergence Truncated Nearest Neighbor (KL-NN) utilizes a symmetric KL-Divergence between the attention map of the prompt point and all attention maps in the attention tensor as distance function. Finally, we also provide a version of M2N2 without flood fill. We observe that the combination of Markov-maps with Flood fill achieves the best results with SD2 as our backbone compared to SD1.1 [36] and ViT-B [32].

SD2 Domain Bias. We notice a significant domain bias of M2N2 on medical data. Evaluating the IoU's for 10 clicks on BraTS [3] (SimpleClick: 87%, M2N2: 70%) and OAIZIB [1] (SimpleClick: 76%, M2N2: 66%) shows significant bias towards natural images.

4.1. Implementation Details

We use the SD2 implementation and weights provided by the Hugging Face transformers package. We don't add noise to the encoded image latent to keep the results deterministic and perform the single denoising step with empty text prompts. Due to the large memory requirements of the attention tensors, we run SD on 16-bit floating-point precision and convert it to 32-bit floating-point for the attention aggregation and further processes in our framework. We implement JBU as described in [19] with two modifications. We change the low-resolution solution sampling from sparse to dense sampling for smoother results and extend the range term to an isotropic Gaussian to better utilize RGB information in the images. We set $\sigma_{\text{spatial}} = 1$ and $\sigma_{\text{range}} = 0.1$ for RGB color values in the range $[0, 1]$. For M2N2 we choose the attention tensors of the size $128 \times 128 \times 128 \times 128$ and the SD time step of 100. We use the temperature $T = 0.65$ together with a relative probability threshold $\tau = 0.3$ to compute the Markov-maps with a maximum of 1000 iterations. By caching the attention tensor and Markov-maps of previous clicks, we observe an average of 0.6 seconds per click (SPC) for an image resolution of 854×480 on a RTX 4090, enabling a near real-time interactive process with the user. For more information on the other backbones please see Sec. 13.

4.2. Ablation Study

We perform extensive ablation studies on the hyperparameters of our segmentation algorithm to demonstrate the impact of each component of M2N2.

Attention Aggregation. Our experiments in Fig. 4 show that higher attention tensor resolution improves NoC significantly for most of our datasets, performing best at a resolution of $128 \times 128 \times 128 \times 128$. This is surprising since it requires an input image size of 1024×1024 which is beyond SD2's training resolution of 768×768[5]. We also evaluate the NoC for the SD time step, which is required for the single denoising step. Time steps greater than 200 increase the NoC which we assume is due to the distribution shift caused by not adding noise to the encoded latent. For the aggregation of attention tensors, we evaluate each attention block individually in Tab. 1 and observe that the attention tensors of up0 and up1 individually achieve significantly better NoC than the other layers. Aggregating up0 and up1 results in the best NoC.

Markov-maps. Lowering the temperature T of the aggregated tensor A gradually improves the NoC in Fig. 4 up to $T = 0.5$. Smaller values of $T < 0.5$ reduce the entropy and therefore require more iterations in the Markov chain, exceeding the maximum number of iterations and causing numerical instabilities. Different settings of the relative probability threshold τ prove to have a relatively low impact on the NoC.

[5]Due to memory requirements we did not test higher resolutions.

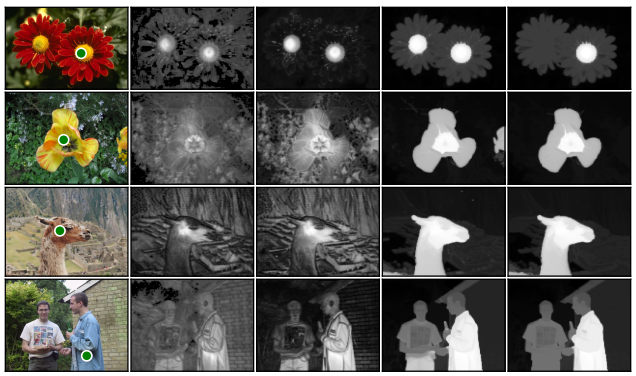


Figure 3. **Comparison of semantic maps**. Each map is generated from a single prompt point. For better comparison, Markov-maps are inverted such that the lowest value is white and the highest value is black.

Truncated Nearest Neighbor. Experiments in Tab. 2 show the contribution of each score function $s_{i,\cdot}(\lambda)$ to the NoC of the total score function $s_i(\lambda)$. In the second row we observe that the function $s_{i,\text{prior}}$, has the lowest impact on the NoC. Removing $s_{i,\text{edge}}$ on the other hand increases NoC90 to 9.99 and is therefore the most important score function.

4.3. Qualitative Results and Limitations of M2N2

Fig. 3 compares our proposed Markov-maps with other semantic maps obtained from a single prompt point. We observe that Markov-maps are less noisy, have clearer semantic boundaries, and have more uniform values within semantically similar regions. Additionally, we notice that Markov-maps nicely reflect a semantic hierarchy due to the segment size ambiguity of a single prompt point. For example, in the first row and right-most column, the two overlapping flowers have a hierarchy of three levels. The first level contains the right flower's ovary, the second contains both flowers together and the final level covers everything. This example also shows the strengths and weaknesses of our flood fill approach. As the ovaries are not overlapping, it enables instance segmentation of the right flower's ovary by suppressing the local minimum of the left flower's ovary. On the other hand, flood fill does not separate the right flower individually, because both flowers are semantically equal and overlapping. Please note that this limitation only applies to Markov-maps of a single point. M2N2 is still able to segment overlapping instances with the truncated nearest neighbor of multiple prompt points. (See Fig. 11)

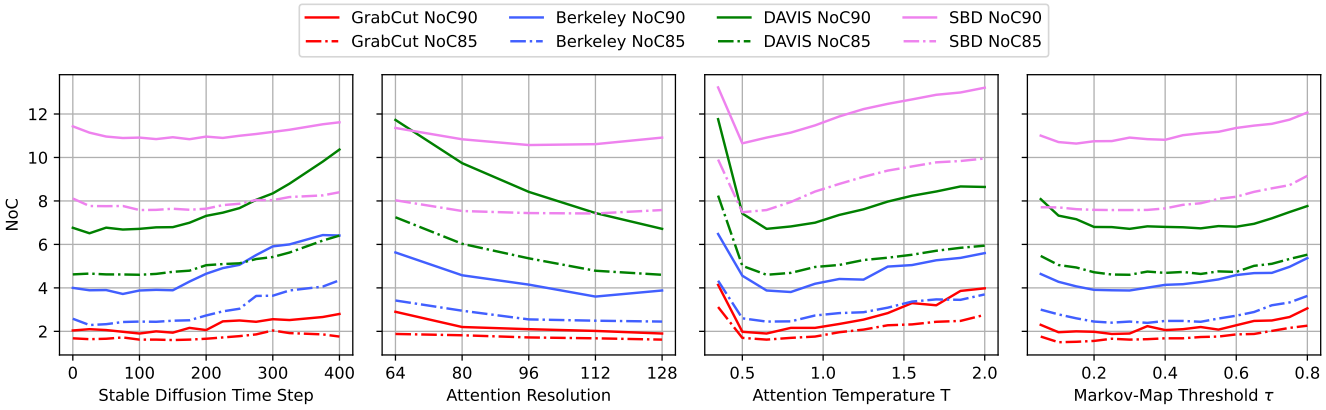


Figure 4. **Impact of the hyperparameters of SD and Markov-map**, respectively, on all four datasets, each represented by a single color. Dashed lines correspond to NoC85, continues lines to NoC90. The graph of SBD is based on a randomly sampled subset of 500 images.

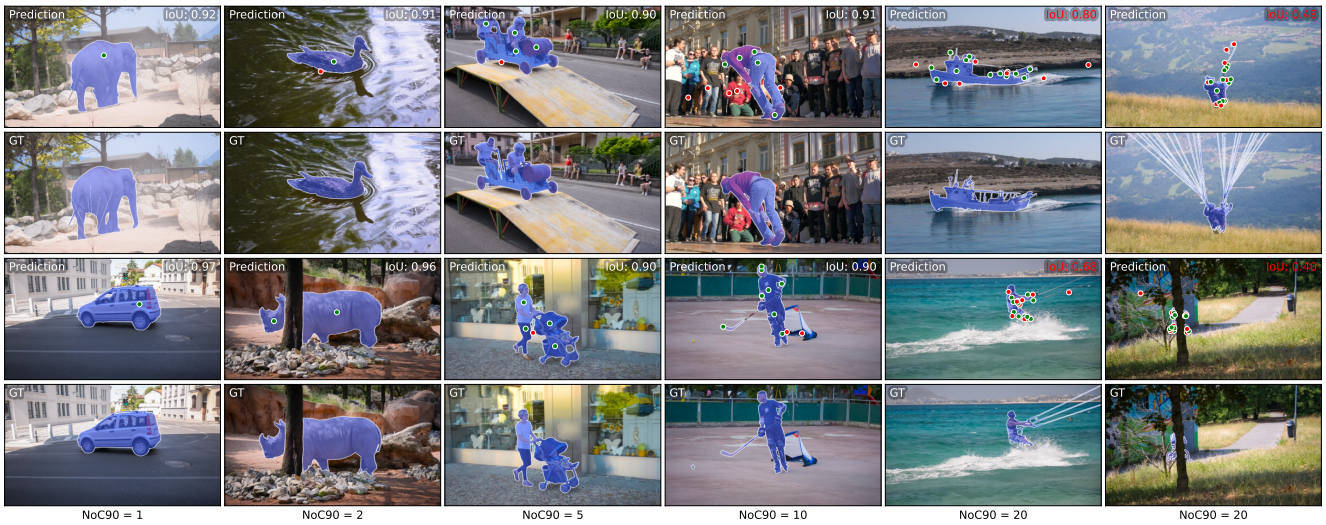


Figure 5. **Segmentation examples on DAVIS [33]**. Each column shows examples slected on the NoC90 value, ranging from easy cases $\mathrm{NoC90} = 1$ on the left to difficult cases $\mathrm{NoC90} = 10$ and failure cases $\mathrm{NoC90} = 20$ on the right. Foreground points are shown in green and background points in red. The bottom right example is especially difficult for M2N2 by only having small thin isolated structures.

Examples from the DAVIS dataset in Fig. 5 show a similar issue in the $\mathrm{NoC90} = 10$ column, where a single dancer is in front of a semantically similar crowd, requiring multiple corrective background points for proper separation. Another limitation of flood fill is that obstructions can split instances into multiple areas. For example, the segmentation of the obstructed rhinoceros requires two points, instead of one. Due to the limited attention resolution, M2N2 faces challenges in segmenting thin and fine structures, as shown in the 2 rightmost columns. The pixel-level truncated nearest neighbor still enables the segmentation of such structures but requires significantly more prompt points to reach the desired accuracy. In general, we observe that M2N2 generates consistent segments with sharp semantic boundaries without being trained on any segmentation labels.

5. Conclusion

We proposed M2N2, a novel method for unsupervised training-free point prompt based interactive segmentation. By interpreting an aggregated self-attention tensor of Stable Diffusion 2 as a Markov transition operator, we generated semantically rich Markov-maps. We showed that Markov-maps have less noise, clearer semantic boundaries, and more uniform values for semantically similar regions. By combining Markov-maps with truncated nearest neighbor, we developed M2N2, which even outperformed state-of-the-art unsupervised training-based methods in most of the datasets. Current limitations are the segmentation of fine structures due to low attention resolution, as well as overlapping or obstructed segments, where M2N2 may require more prompt points.

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